Chapter 6.2
Step by Step Pulse

회로이론

HRI 연구실 김동한

Human-Robot Interaction Laboratory

KYUNG HEE UNIVERSITY

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단계별 기법

>> 단계별 기법

모든 페이지에 해당 폰트로 통일(주제목, 소제목, 수식 제외), 맑은 고딕
ppt 홈 -> 서식 복사 두번 클릭하면 여러 곳 적용 가능

- 이 기법은 알려진 해의 형식을 사용한다.그렇지만 기본 회로해석 툴을 이용하여 상수 K_1,K_2, τ 를 구하므로 미분방정식 모델을 사용하지 않는다.

$$x(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$$

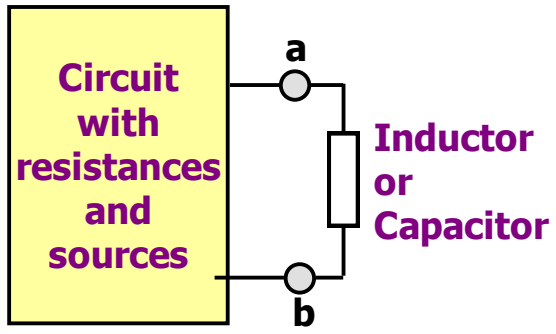
K_1 은 해당 변수의 정상상태 값이며, 회로를 정상상태에서 해석함으로써 결정할 수 있다.

$x(0+)$ 는 초기 조건이며, 상수 K_1, K_2를 계산하기 위한 두 번째 방정식을 제공한다.

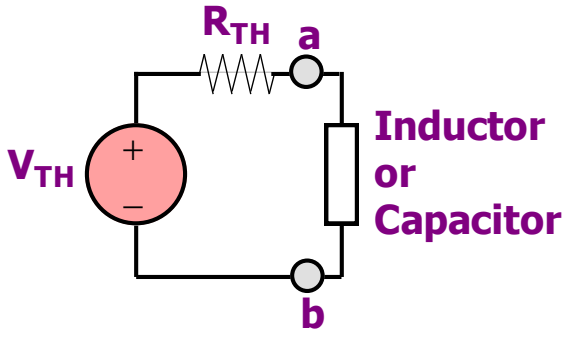
τ는 시상수이며, 에너지 저장 소자 양단에서 테브난 등가를 사용하여 결정할 수 있다.

단계별 기법

>> 시정수 결정: 일반적인 기법



Representation of an arbitrary circuit with one storage element

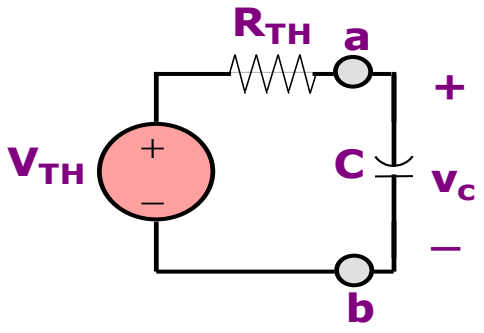


Capacitive Circuit $\tau = R_{TH}C$

Inductive Circuit $\tau = \frac{L}{R_{TH}}$

단계별 기법

>> 시정수 결정: 일반적인 기법



Case 1.1
Voltage across capacitor

- **KCL 노드 a**

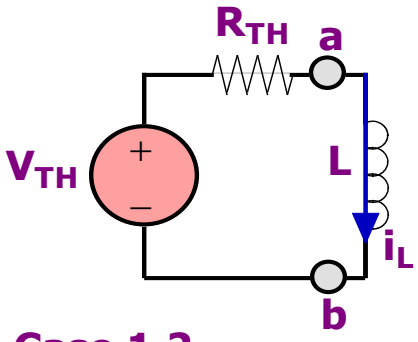
$$i_c + i_R = 0$$

$$i_c = C\frac{dv_c}{dt}$$

$$i_R = \frac{v_c - v_{TH}}{R_{TH}}$$

$$C\frac{dv_c}{dt} + \frac{v_c - v_{TH}}{R_{TH}} = 0$$

$$R_{TH} = C\frac{dv_c}{dt} + v_c = v_{TH}$$



Case 1.2
Current through inductor

- **KVL 사용**

$$i_c + i_R = 0$$

$$v_R = R_{TH} i_L$$

$$v_L = L\frac{di_L}{dt}$$

$$L\frac{di_L}{dt} + R_{TH} i_L = v_{TH}$$

$$\left(\frac{L}{R_{TH}}\right)\frac{di_L}{dt} + i_L = \frac{v_{TH}}{R_{TH}} = i_{sc}$$

단계별 기법

단계

- **STEP 1. 해의형식**

$$x(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, \mathrm{t} > 0$$

$$K_1 = x(\infty);\ K_1 + K_2 = x(0+)$$

DETERMINE $\boldsymbol{x}(0+)$

- **STEP 2: 스위칭 전의 정상상태회로를 그리고 캐패시터 전압 혹은 인덕터 전류를 구한다.**

- **STEP 3: 0+에서의 회로를 그린다. 캐패시터는 전압원처럼 인덕터는 전류원처럼 작동. t=0+에서의 변수를 구한다.**

DETERMINE $\boldsymbol{x}(\infty)$

- **STEP 4: 스위칭 이후 정상상태 회로를 그리고 정상상태에서의 변수를 구한다.**

- **STEP 5: 시정수를 구한다.**

$$\tau = R_{TH}C \quad circuit\ with\ one\ capacitor$$

$$\tau = \frac{L}{R_{TH}} \quad circuit\ with\ one\ indictor$$

- **STEP 6: 상수 K1, K2를 구한다.**

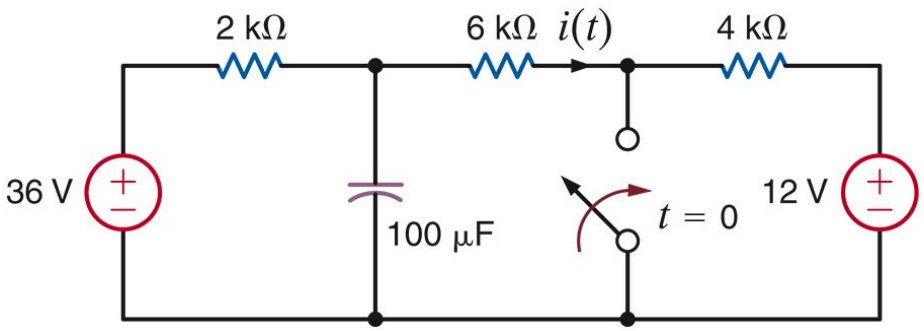
$$K_1 = x(\infty), K_1 + K_2 = x(0+)$$

단계별 기법

>> 학습 예제

STEP 3 : Determine $i(0+)$

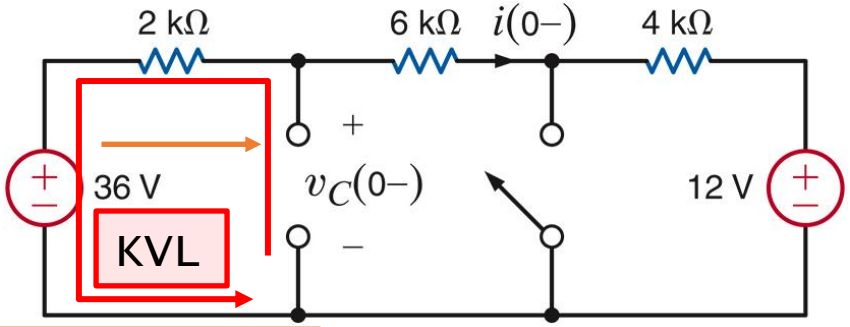
- t>0 일때 i(t)를 구하라.



STEP 1: $i(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP 2: 커패시터 양단의 초기 전압

스위칭 이전의 정상상태 회로 사용



$$i = \frac{24V}{12K\Omega} = 2mA$$

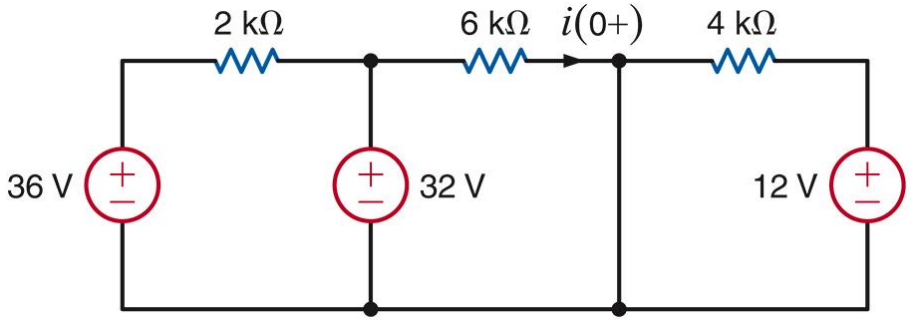
$$= i(0) -$$

$$v_c(0+) = v_c(0-)$$

$$v_c(0-) = 36V - (2mA)(2k) = 32V$$

STEP 3: $i(0+)$ 결정

t=0+에 맞는 회로 사용.
캐패시터는 전원처럼 작동



$$i(0+) = \frac{32V}{6k} = \frac{16}{3} mA$$

유도성 회로의 유의점

- 단계 2에서 인덕터 전류의 초기 값 구함
- t=0+에 대해 인덕터를 전류원으로 대체

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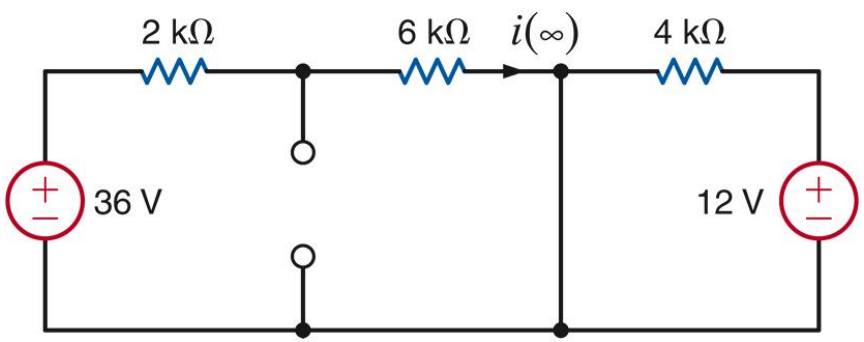
단계별 기법

>> 학습 예제

STEP 4: $i(\infty)$ 결정

스위칭 이후의 정상상태 회로 사용

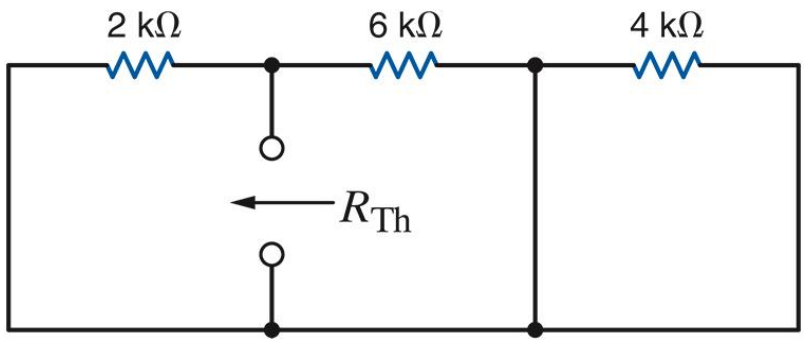


36V 전원에 의해 생성되는 전류

$$i(\infty) = \frac{36}{8} = \frac{9}{2} mA$$

STEP 5 : 시정수 결정

$$Capacitive\ circuit:\ \tau = R_{TH}C$$



$$R_{TH} = 2k||6k = 1.5k\Omega \quad C=100\mu F$$

$$\tau = (1.5 \times 10^3\Omega)(100 \times 10^{-6}F) = 0.15s$$

단계별 기법

>> 학습 예제

STEP 6 : Determine $\boldsymbol{K}_1, \boldsymbol{K}_2$

$$i(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$$

$$i(0+) = \frac{16}{3} mA = K_1 + K_2$$

$$i(\infty) = \frac{9}{2} mA = K_1$$

$$\therefore K_2 = \frac{16}{3} - \frac{36}{8} = \frac{5}{6}$$

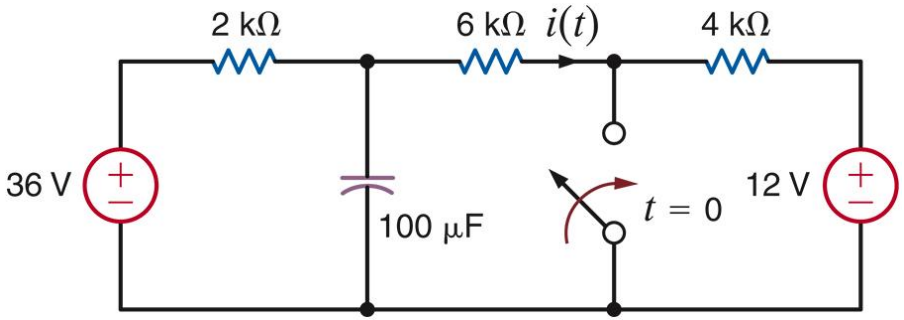
FINAL ANSWER

$$i(t) = \frac{9}{2} + \frac{5}{6} e^{-\frac{t}{0.15}}, t > 0$$

유의: 유동성 회로에서

$$\tau = \frac{L}{R_{TH}}$$

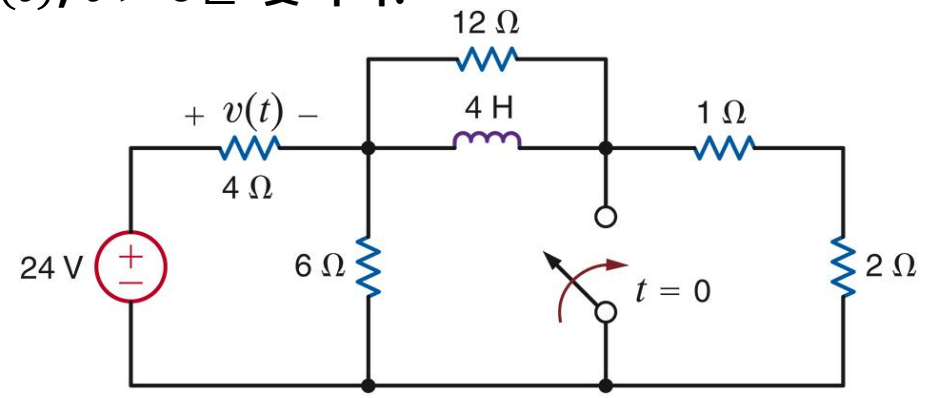
- **원 회로**



단계별 기법

>> 학습 예제

- $v(t)$, $t > 0$를 찾아라.



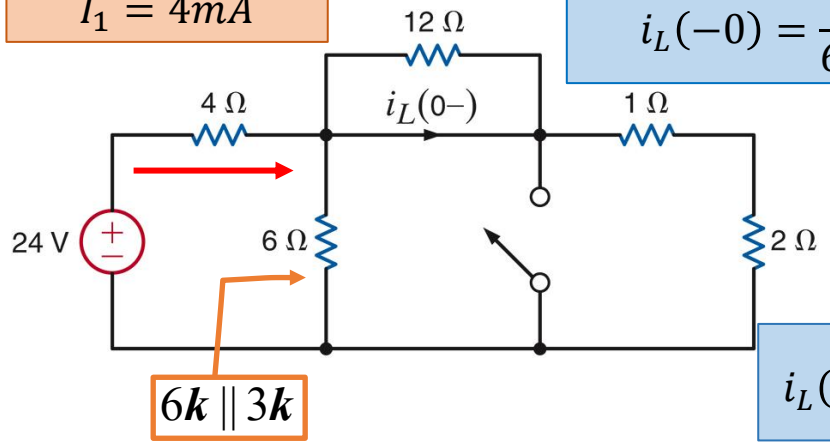
STEP 1: $v(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP 2: 인덕터 초기 전류 계산

스위칭 이전의 정상상태 회로 사용

$I_1 = 4mA$

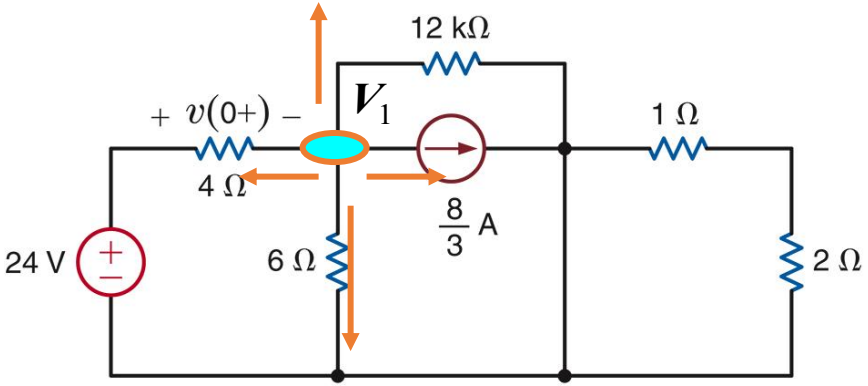
$$i_L(-0) = \frac{6}{6+3} I_1$$



$$i_L(0-) = \frac{8}{3} mA$$

STEP 3: $v(0+)$ 결정

- t=0+에서의 회로 사용.
- 인덕터는 전류원으로 대체



$$\frac{V_1 - 24}{4} + \frac{V_1}{6} + \frac{V_2}{12} + \frac{8}{3} = 0$$

$$V_1 = \frac{20}{3} V$$

$$v(0+) = 24V - V_1 = \frac{52}{3} V$$

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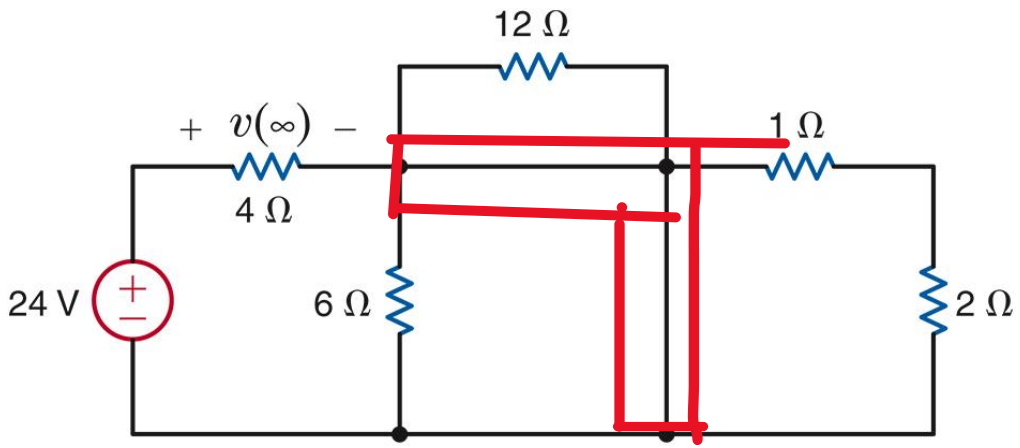
단계별 기법

>> 학습 예제

STEP 4: v(∞) 결정

스위칭 이후의 정상상태 회로 사용



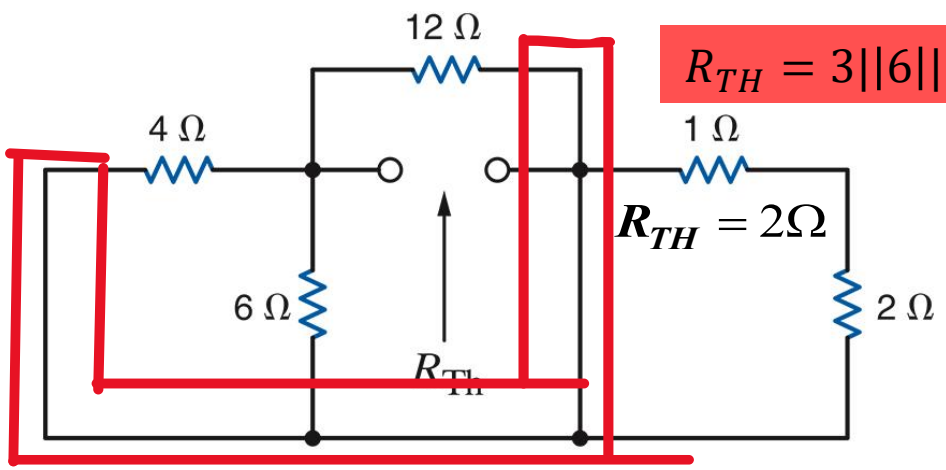
$$v(\infty) = 24[V]$$

STEP 5 : 시정수 결정

$$\tau = \frac{4H}{2\Omega} = 2s$$

Inductive Circuit: $\tau = \frac{L}{R_{TH}}$

$R_{TH} = 3||6||12$



단계별 기법

>> 학습 예제

STEP 6 : K_1, K_2 결정

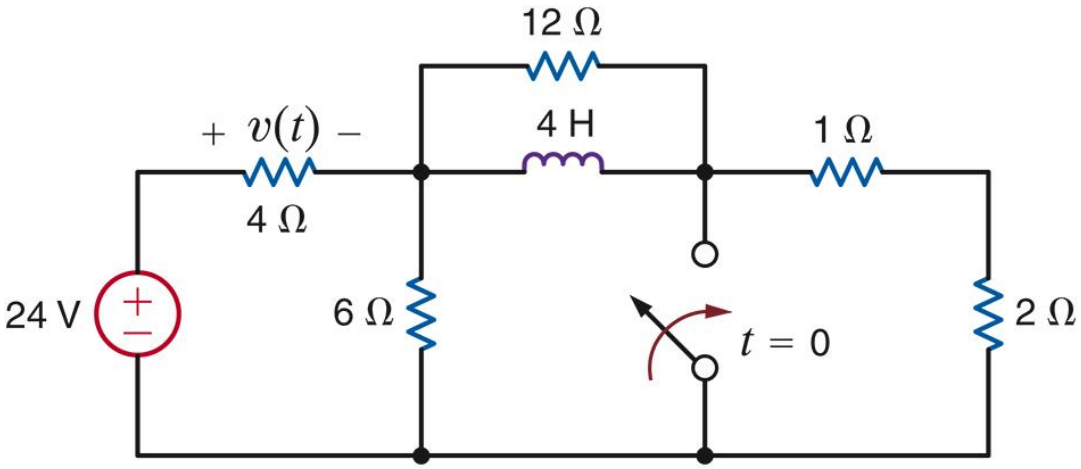
$$K_1 = v(\infty) = 24V$$

$$v(0+) = \frac{52}{3} = K_1 + K_2$$

$$K_1 = \frac{52}{3} - 24 = -\frac{20}{3}V$$

$$ANS: v(t) = -\frac{20}{3} + 24e^{-\frac{t}{2}}, t > 0$$

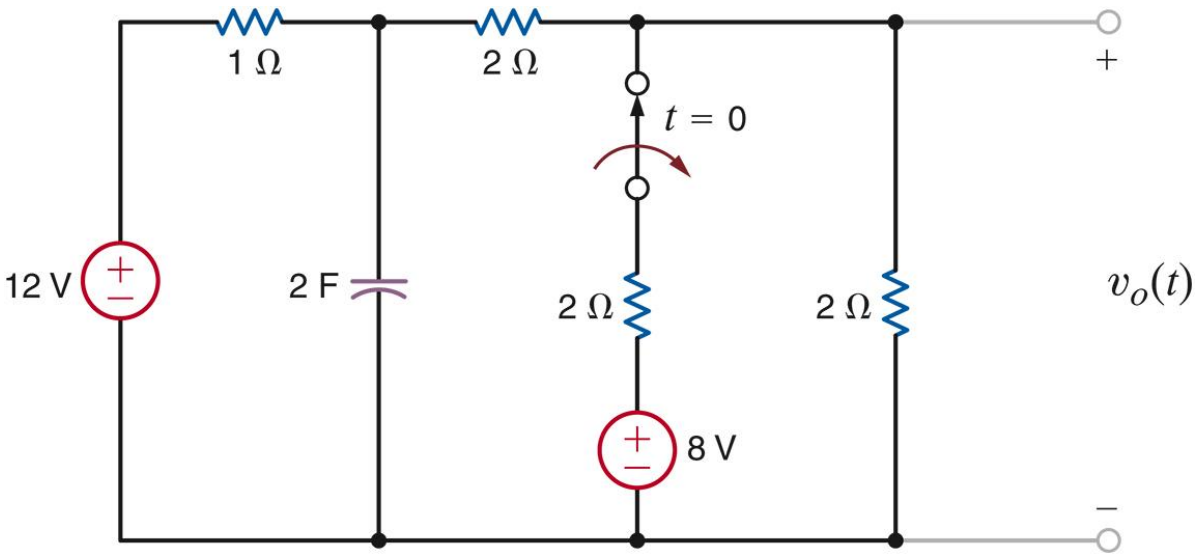
- 원 회로



단계별 기법

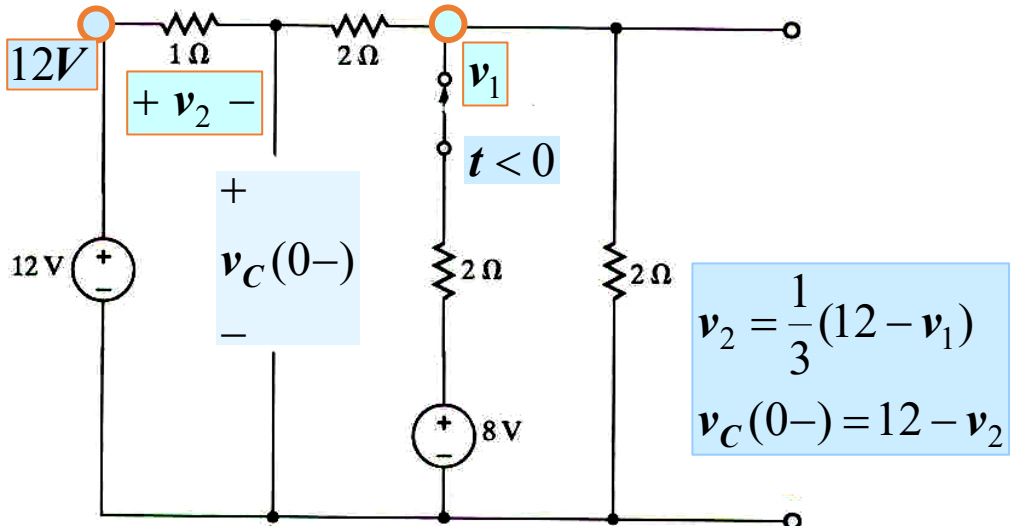
>>추가학습

- t > 0 일 때 $v_0(t)$ 를 구하라.



STEP 1: $v(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP2: 2단계: 커패시터의 초기 전압



KCL v_1 : $\frac{v_1 - 12}{3} + \frac{v_1 - 8}{2} + \frac{v_1}{2} = 0$

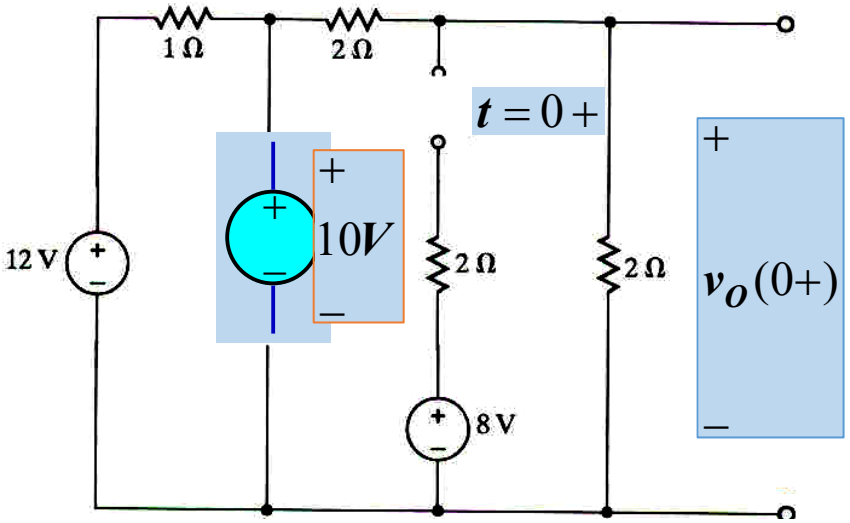
$$8v_1 - 48 = 0 \Rightarrow 6V$$

$$v_2 = 2V \Rightarrow v_c(0-) = v_c(0+) = 10V$$

단계별 기법

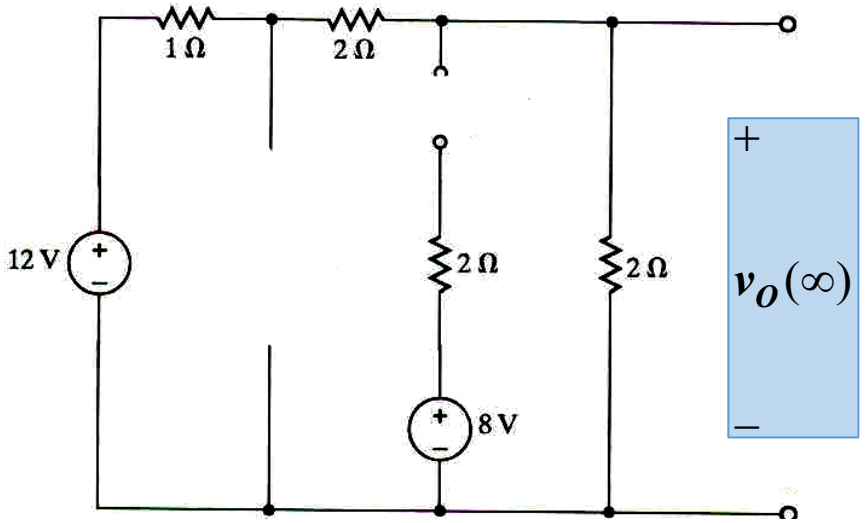
>>추가학습

STEP3: $v_0(0+)$ 결정



$$v_0(0+) = \frac{2}{2+2}(10) = 5V$$

STEP4 : $v_0(\infty)$ 결정



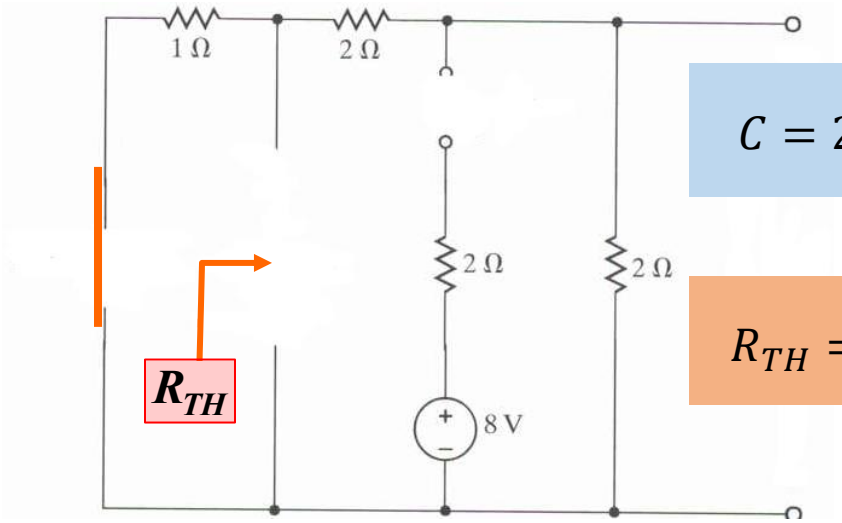
$$v_0(0+) = \frac{2}{5}(12) = \frac{24}{5}V$$

단계별 기법

>>추가학습

STEP 5 : 시정수 결정

Capacitive Circuit : $\tau = R_{TH}C$



$$C = 2F \Rightarrow \tau = \frac{8}{5}s$$

$$R_{TH} = 1||4 = \frac{4}{5}\Omega$$

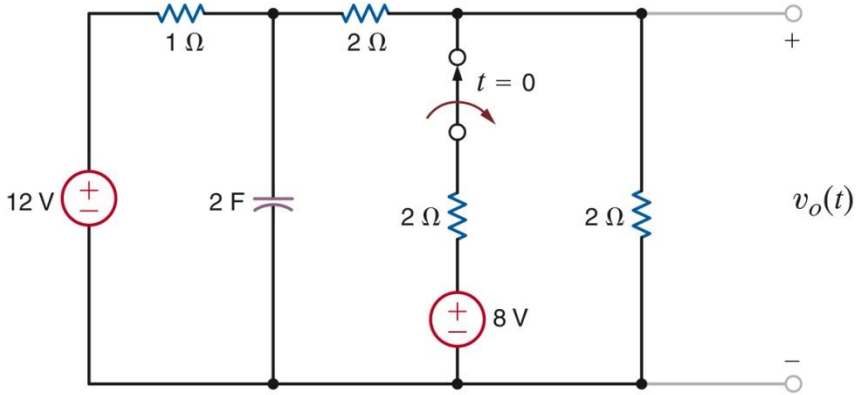
STEP 6 : K_1, K_2 결정

$$K_1 = v_0(\infty) = \frac{24}{5}V$$

$$v_0(0+) = 5V = K_1 + K_2 \Rightarrow K_2 = \frac{1}{5}V$$

$$v_0(t) = \frac{24}{5} + \frac{1}{5}e^{-\frac{t}{8/5}}V; t > 0$$

- 원 회로



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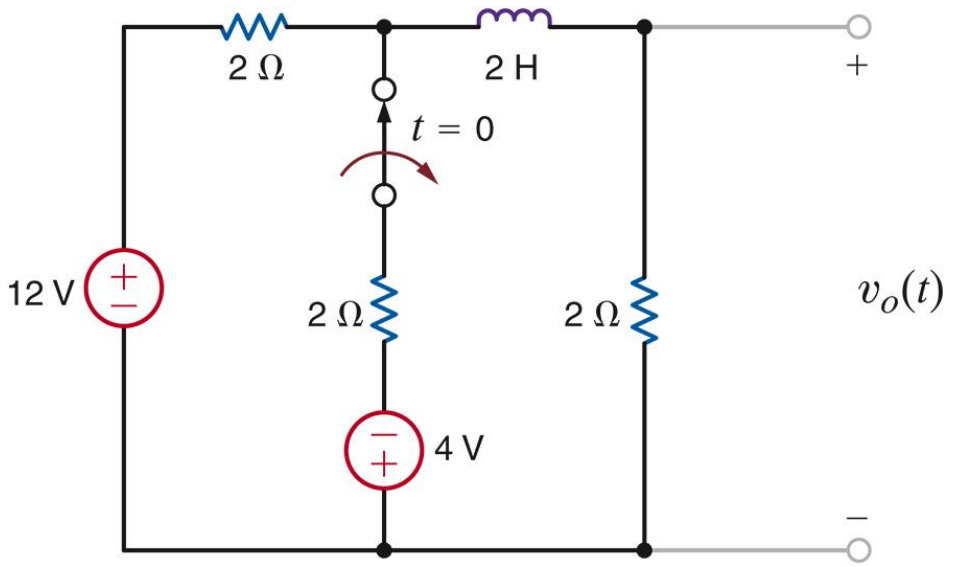
단계별 기법

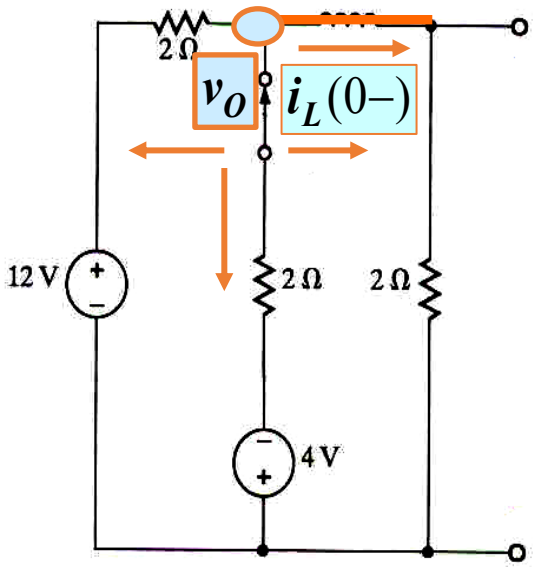
>>추가학습

- t>0일 때 $v_0(t)$를 구하라.



STEP 1: $v_0(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP 2 : 인덕터의 초기 전류



$$\frac{v_o - 12}{2} + \frac{v_o - (-4)}{2} + \frac{v_0}{2} = 0$$

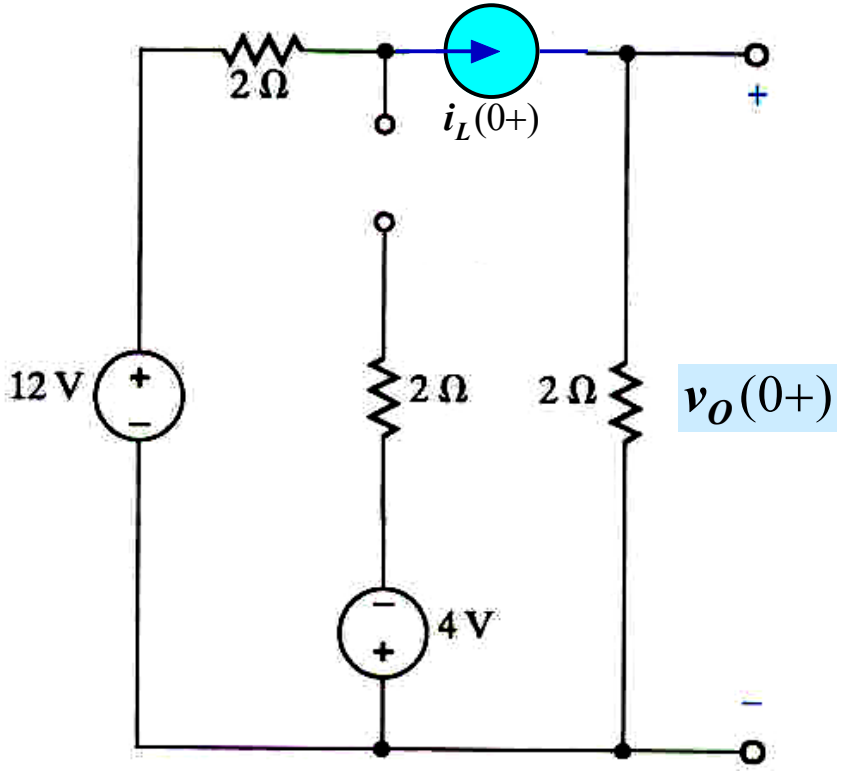
$$v_0 = \frac{8}{3} V$$

$$i_L(0-) = i_L(0+) = \frac{4}{3} A$$

단계별 기법

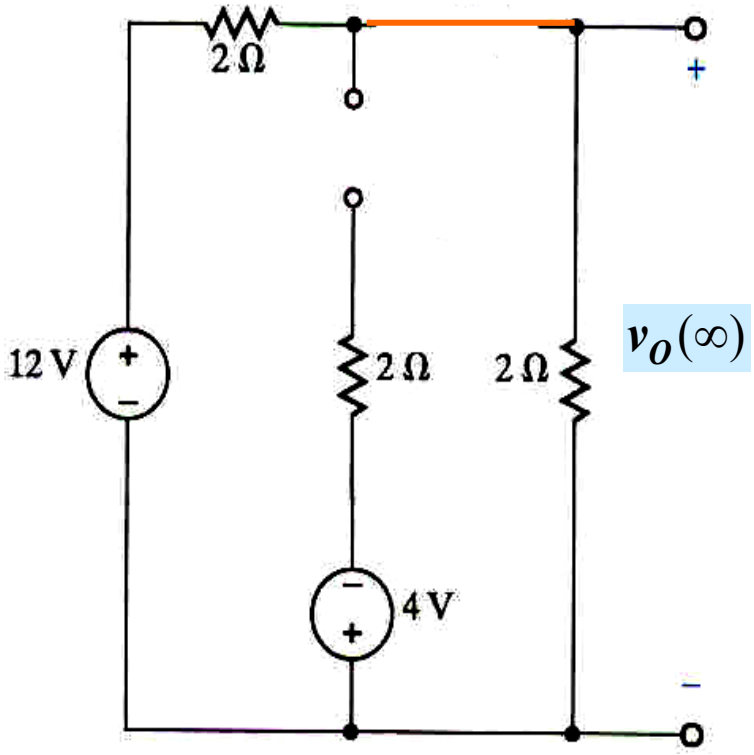
>>추가학습

STEP3: $v_0(0+)$ 결정



$$v_0(0+) = 2i_L(0+) = \frac{8}{3}V$$

STEP4 : $v_0(\infty)$ 결정

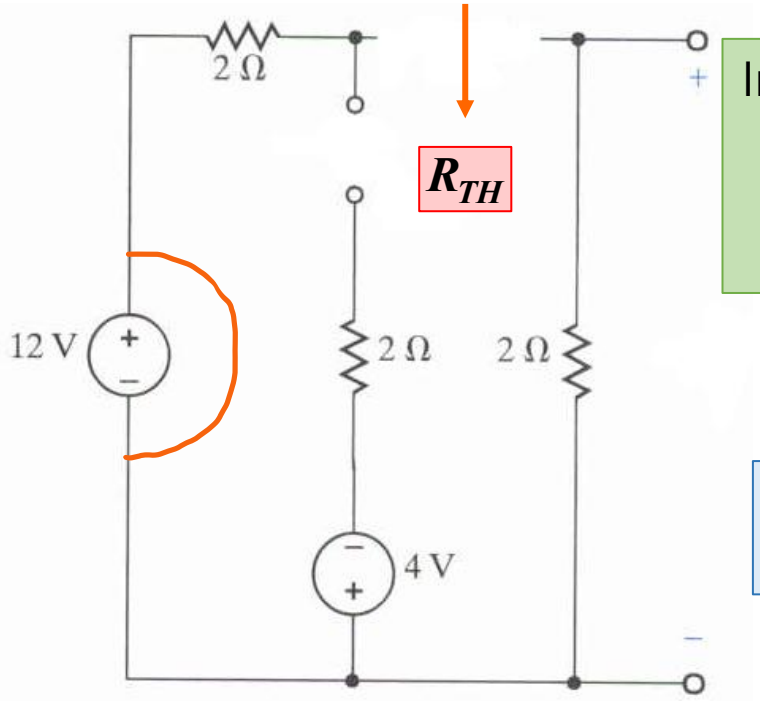


$$v_0(0+) = \frac{2}{2+2}(12) = 6V$$

단계별 기법

>>추가학습

STEP 5 : 시정수 결정



Inductive Circuit

$$\tau = \frac{L}{R_{TH}}$$

$$R_{TH} = 4\Omega$$

$$\tau = \frac{2}{4} = 0.5s$$

STEP 6 : K_1, K_2 결정

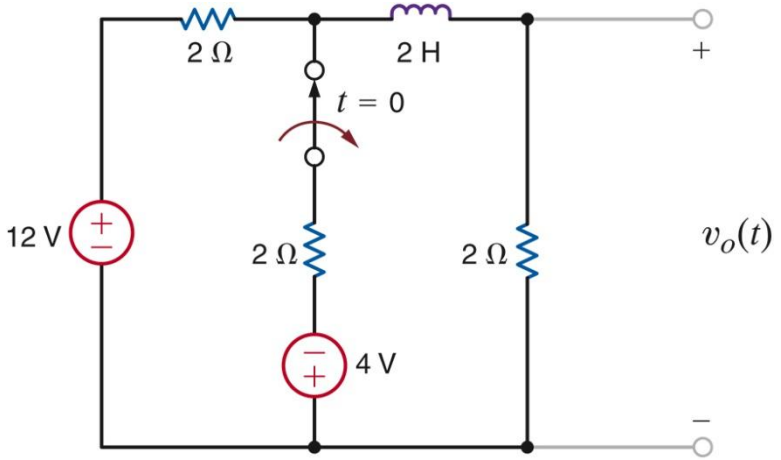
$$K_1 = v_0(\infty) = 6V$$

$$v_0(0+) = \frac{8}{3} = K_1 + K_2$$

$$K_2 = \frac{8}{3} - 6 = -\frac{10}{3}V$$

$$ANS: v_o(t) = 6 - \frac{10}{3}e^{-\frac{t}{0.5}}, \qquad t > 0$$

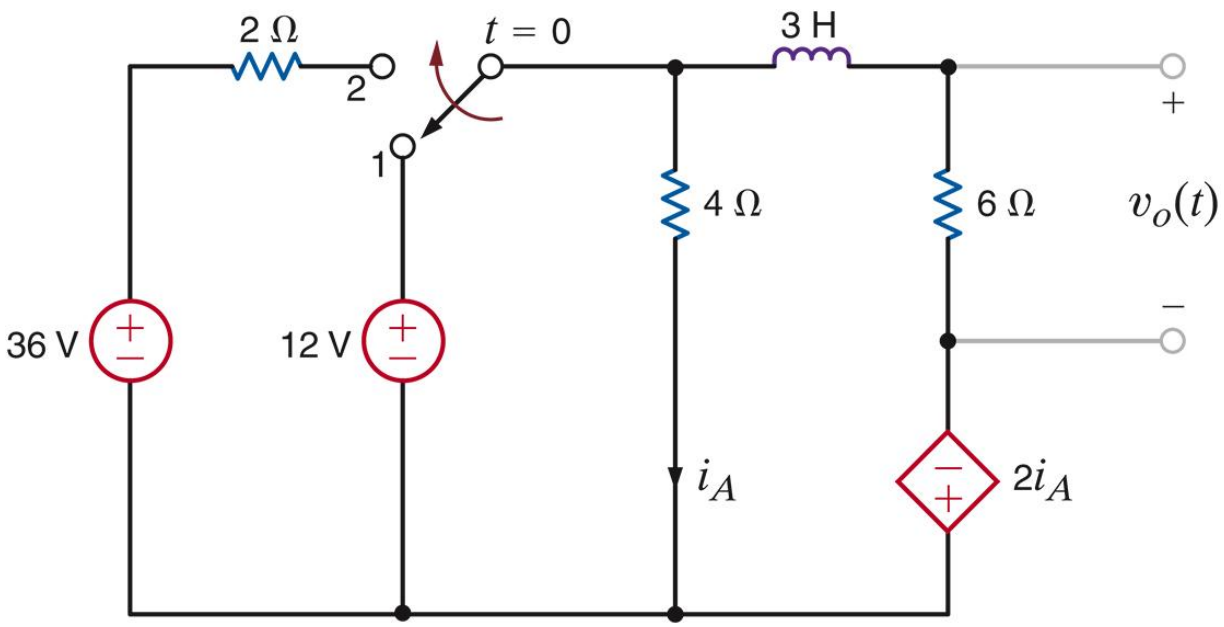
- 원 회로



단계별 펄스

>> 학습 예제

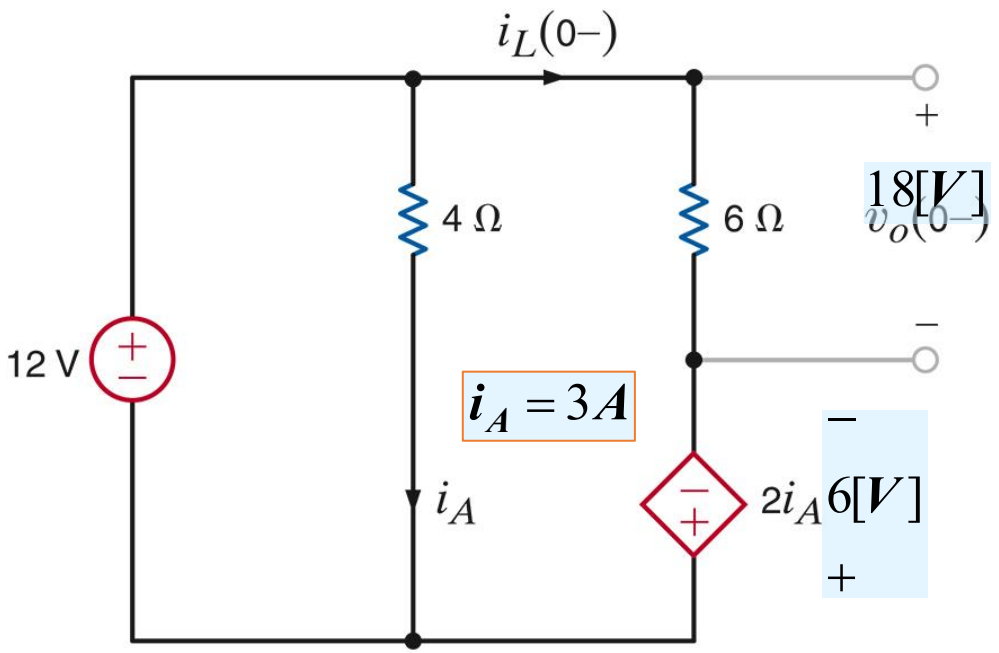
- **t>0일 때 $v_0(t)$를 구하라.**



STEP 1: $v_0(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP2 : $i_L(0+)$ 결정

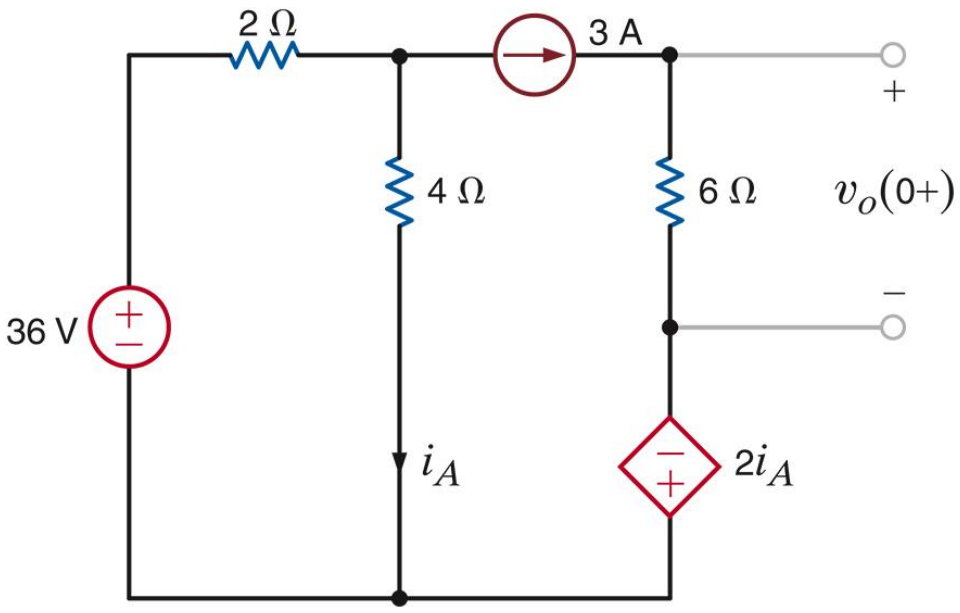
$$i_L(0-) = i_L(0+) = 3A$$



단계별 기법

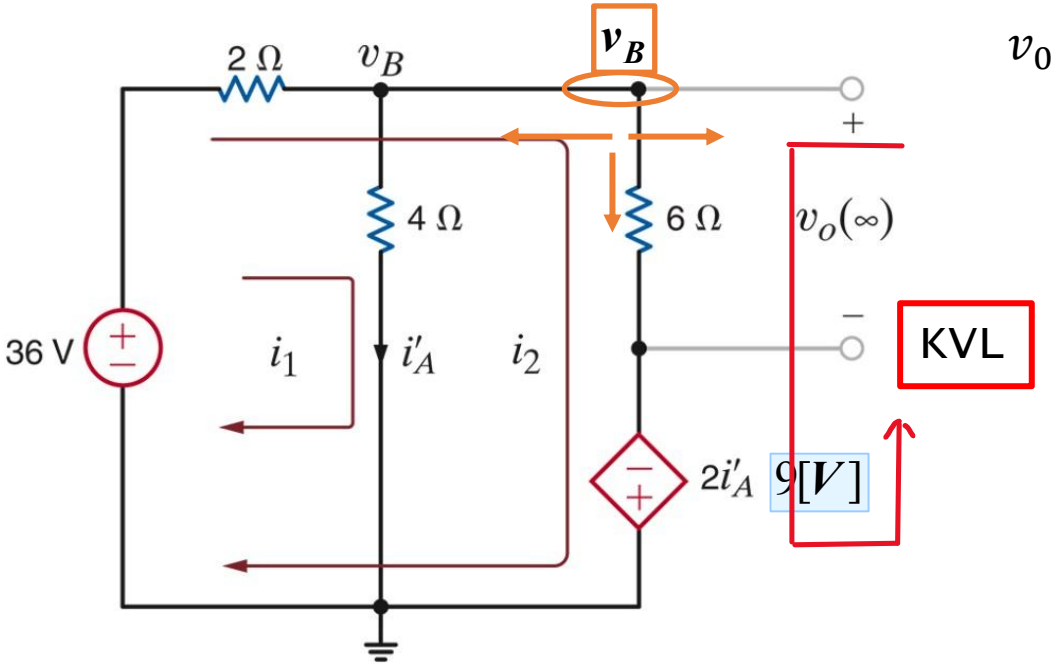
>> 학습 예제

STEP3: $v_0(0+)$ 결정



$$v_0(0+) = 18V$$

STEP4 : $v_0(\infty)$ 결정



$$v_0(\infty) = 27V$$

$$\frac{v_B - 36}{2} + \frac{v_B}{4} + \frac{v_B - (-2i'_A)}{6} = 0$$

$$11v_B + 4i'_A = 4.5\ A$$

$$v_B = 18V, i'_A = 4.5A$$

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단계별 기법

학습예제

STEP 5 : 시정수 결정

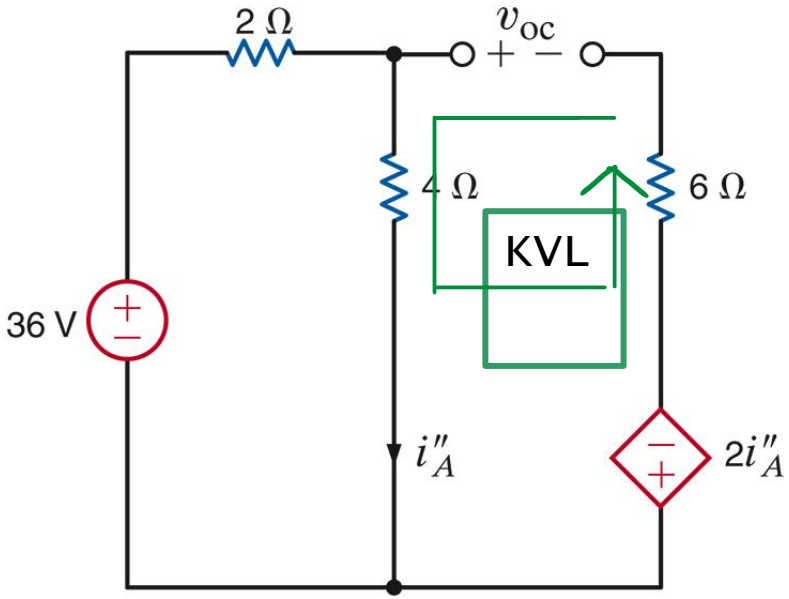
Inductive circuit

$$\tau = \frac{L}{R_{TH}}$$

Circuit with dependent sources

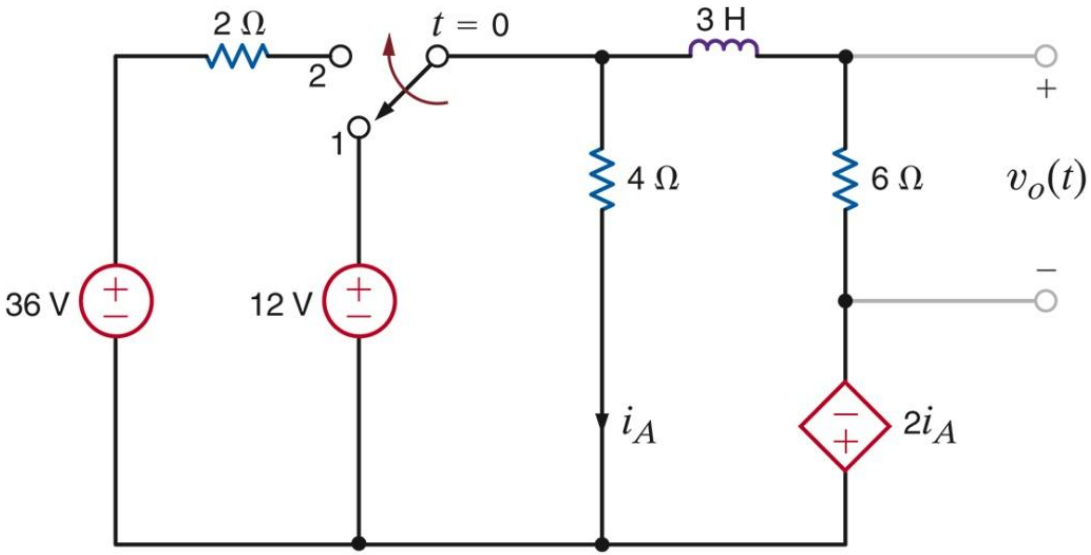
$$R_{TH} = \frac{v_{OC}}{i_{SC}}$$

- 개방 회로 전압



$$v_{OC} = 24 + 12 = 36V$$

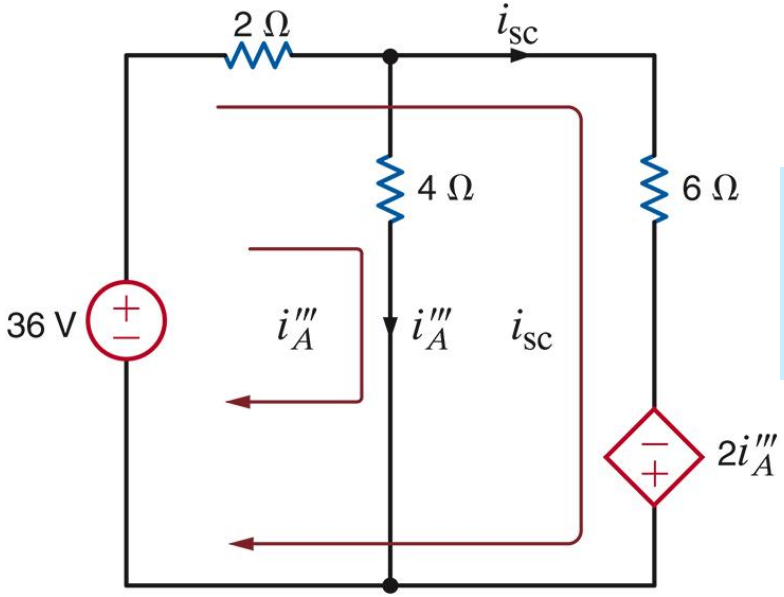
- 원 회로



단계별 기법

>> 학습예제

- 단락회로 전류



$$i_A''' = i_1$$
$$i_{SC} = i_2$$

$$36 = 2(i_1 + i_2) + 4i_1$$
$$36 = 2(i_1 + i_2) + 6i_2 - 2i_A''$$

$$i_{SC} = \frac{36}{8} A$$

$$8\Omega = R_{TH} <= \begin{cases} v_{oc} = 36V \\ i_{sc} = \dfrac{36}{8} A \end{cases}$$

$$L = 3H \Rightarrow \tau = \frac{3}{8} s$$

STEP 6 : K_1, K_2 결정

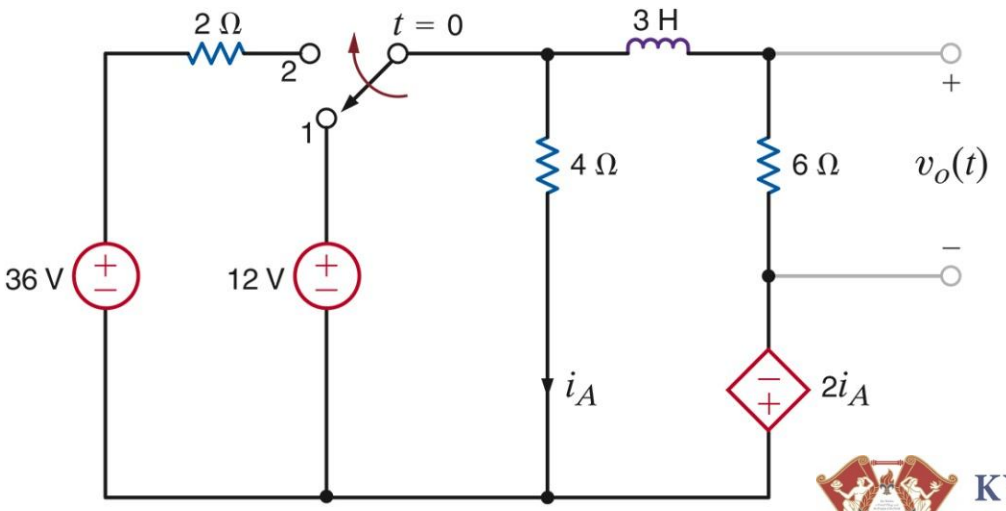
$$v_0(\infty) = 27 = K_1$$

$$v_0(0+) = 18 = K_1 + K_2 \Rightarrow K_2 = -9V$$

$$ANS: \ v_0(t) = 27 - 9e^{-\frac{t}{3/8}}, \qquad t > 0$$

유의: 유도성의 경우, 단락회로 전류를 구하기 위한 회로는 $\mathbf{v_0}(\infty)$ 를 구하기 위한 회로와 같다.

- 원 회로



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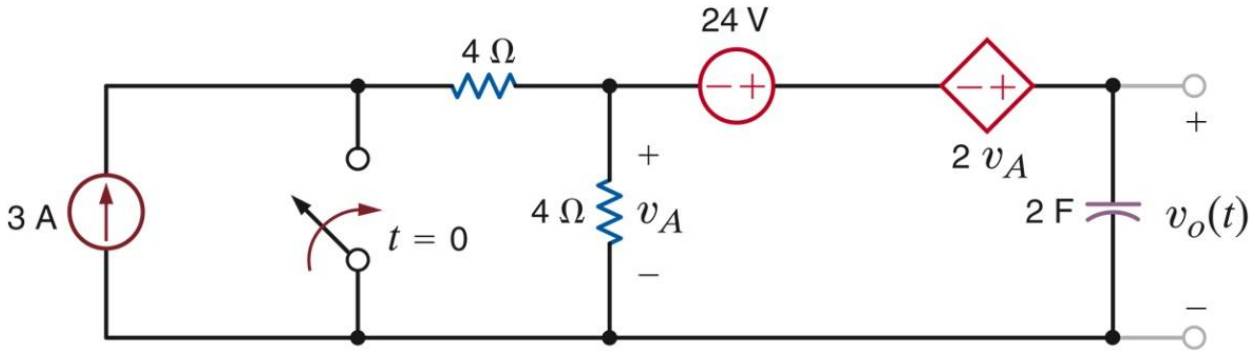
단계별 펄스

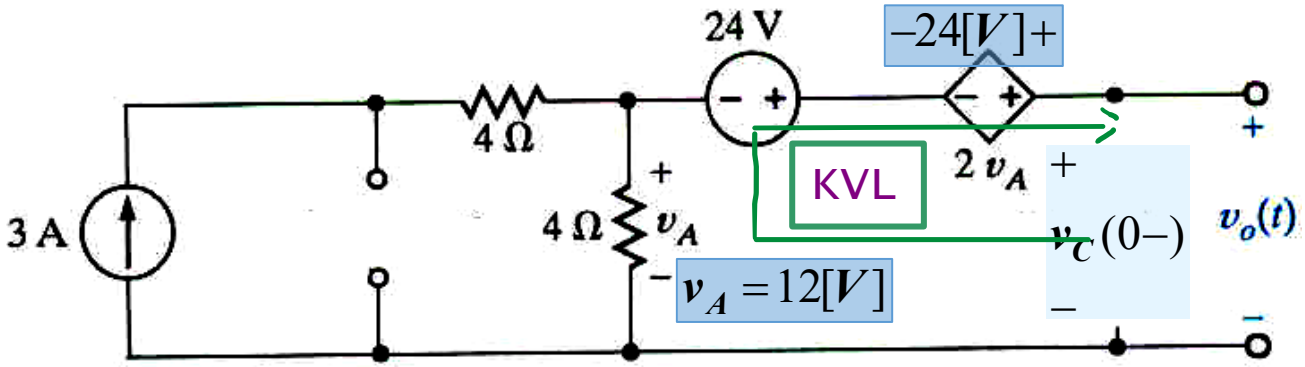
≫ 추가학습

- t<0 일때 $v_0(t)$를 구하라.



STEP 1: $v_0(t) = K_1 + K_2 e^{-\frac{t}{\tau}}, t > 0$

STEP2 : $t = 0+$ 에서의 커패시터 전압 구하기



$$v_c(0-) = 24 + 24 + 12 = 60V$$

$$v_c(0+) = v_c(0-)$$

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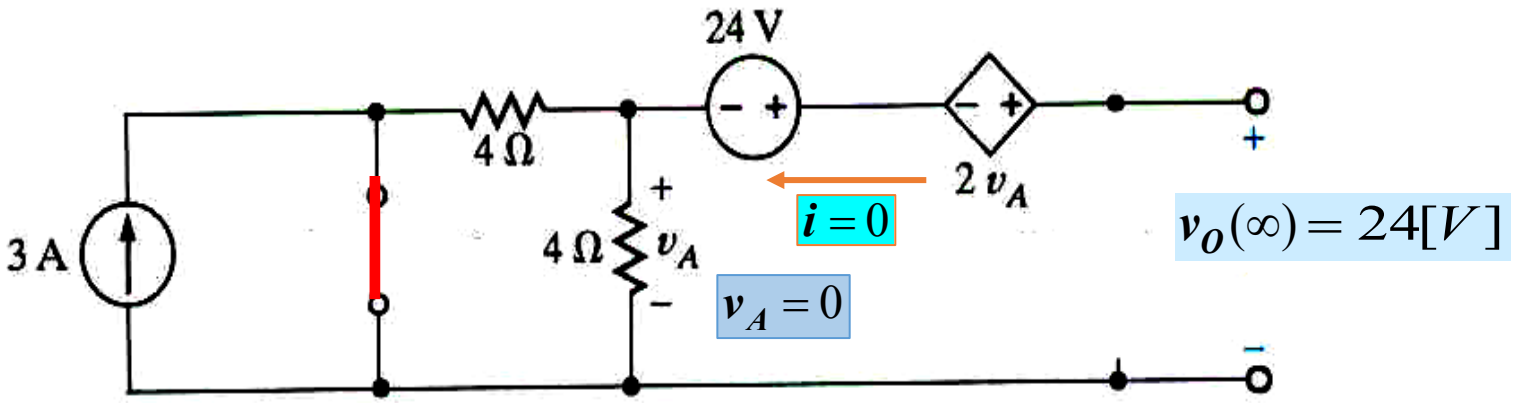
단계별 기법

≫ 추가학습

STEP3: $v_0(0+)$ 결정

$$v_0 = v_c \Rightarrow v_0(0+) = 60V$$

STEP4 : $v_0(\infty)$ 결정



$v_O(\infty) = 24[V]$

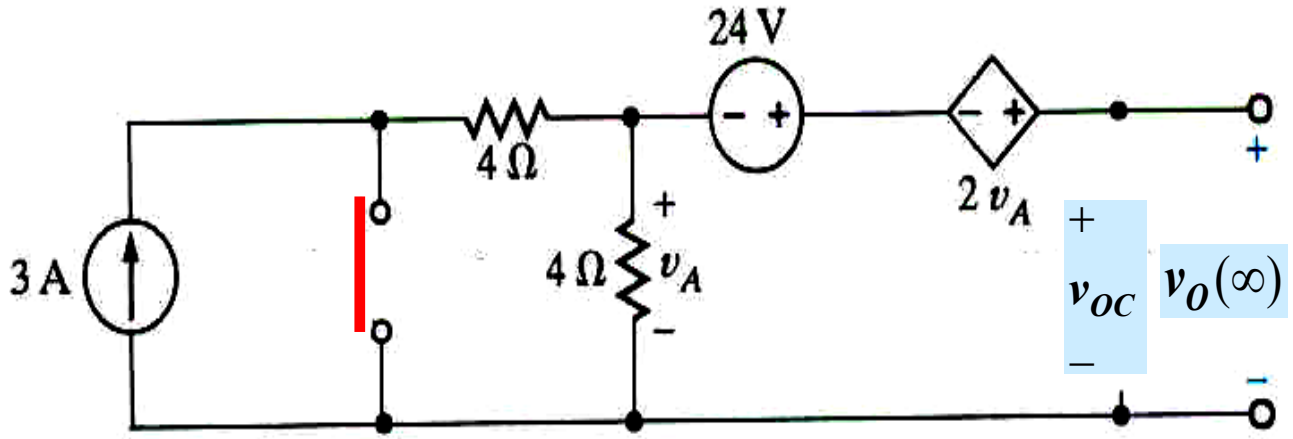
단계별 기법

>>추가학습

STEP 5 : 시정수 결정

Capacitive circuit ⇒ $\tau = R_{TH}C$ $\qquad R_{TH} = \dfrac{v_{oc}}{i_{sc}}$

- 개방회로 전압

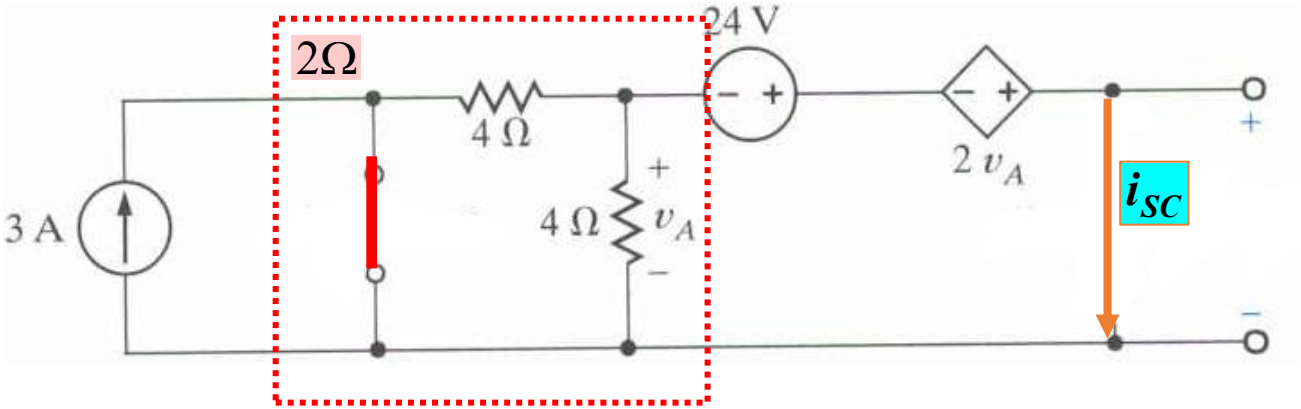


$$v_{oc} = v_0(\infty) = 24V$$

단계별 기법

≫ 추가학습

- 단락회로 전류



$$2i_{sc} - 24 - 2v_A = 0$$

$$v_A = -2i_{sc}$$

$$i_{sc} = 4A$$

$$R_{TH} = \frac{24}{4} = 6\Omega$$

$$\tau = 6\Omega \times 2F = 12s$$

STEP 6 : K_1, K_2 결정

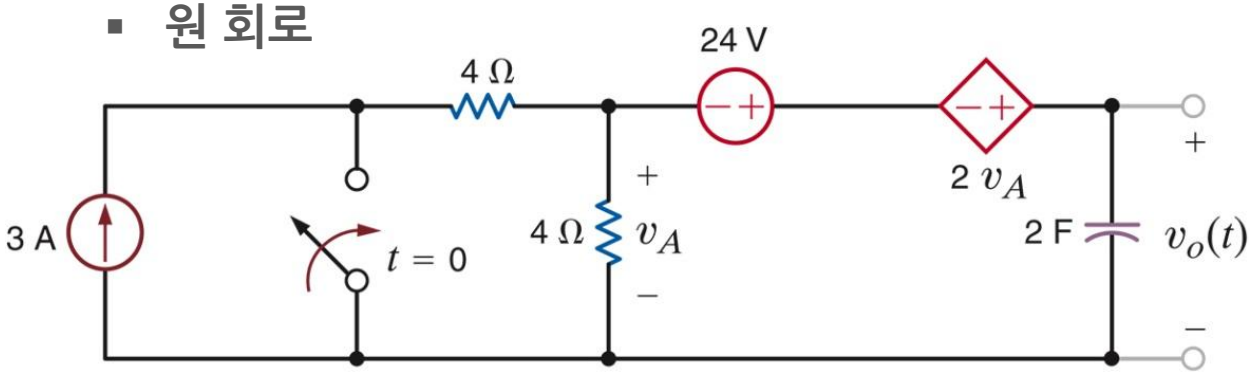
$$K_1 = v_0(\infty) = 24$$

$$v_0(0+) = 60 = K_1 + K_2$$

$$K_2 = 36V$$

$$ANS: v_0(t) = 24 + 36e^{-\frac{t}{12}}, t > 0$$

- 원 회로

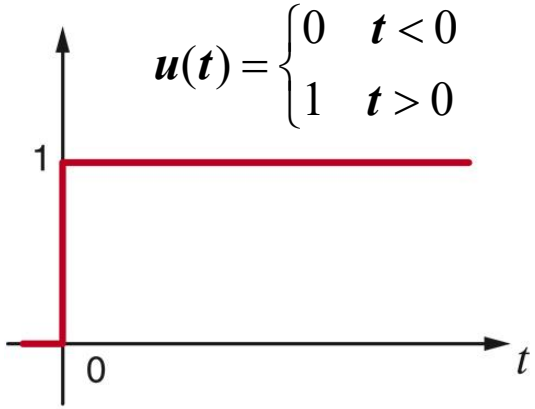


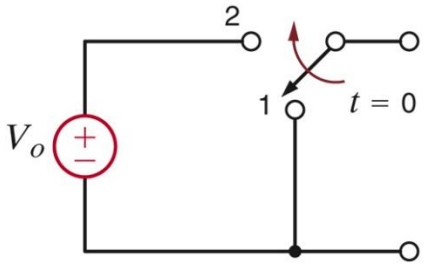
펄스 응답

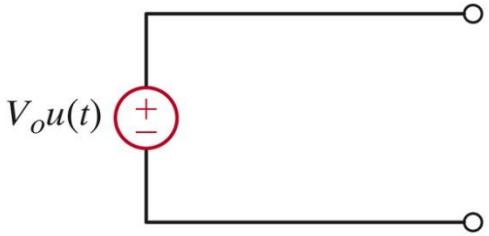
펄스 응답

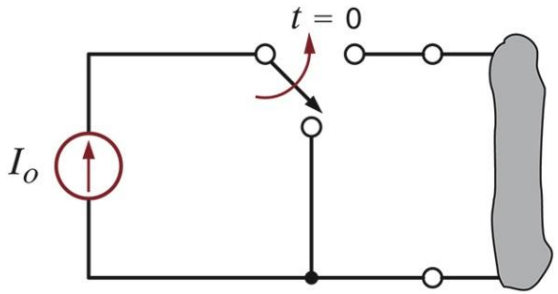
- **어떤 시스템에 펄스 신호를 입력했을 때 시스템의 출력단에서 나타나는 반응 또는 특성**

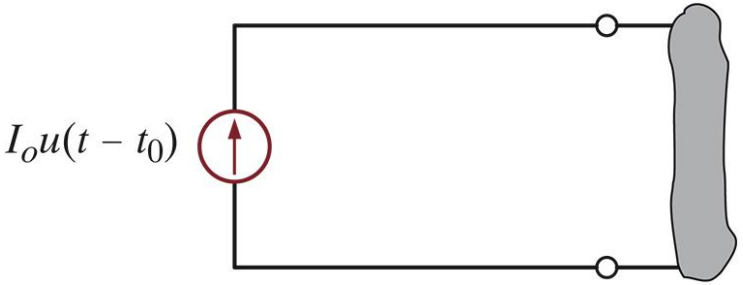
  특이점 함수 계열 입력에 대한 회로의 응답을 학습한다.

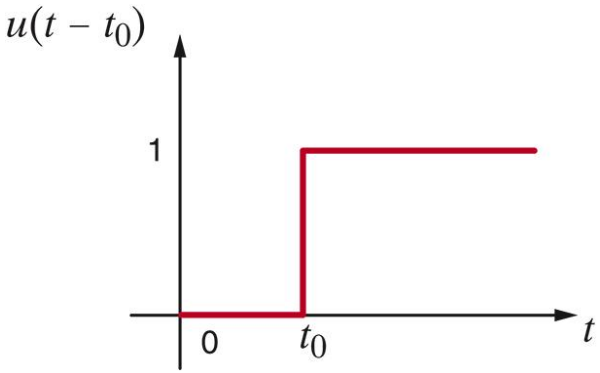






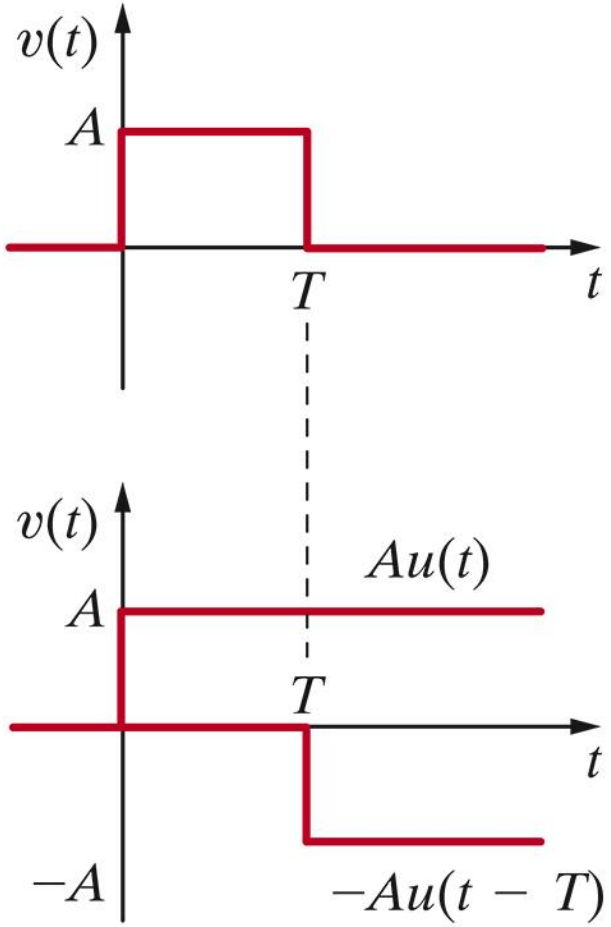






펄스 응답

펄스 신호



계단파의 합으로 형성된 펄스

펄스 응답

>> 0이 아닌 초기 값 및 반복 스위칭

$$\tau \frac{dx}{dt} + x = f_{TH}; x(t_0 +) = x_0$$

$$x(t) = e^{-\frac{t-t_0}{\tau}} x(t_0) + \frac{1}{\tau} \int_{t_0}^{t} e^{-\frac{t-x}{\tau}} f_{TH}(x) dx$$

$$x(t) = K_1 + K_2 e^{-\frac{t-t_0}{\tau}}; t \geq t_0$$

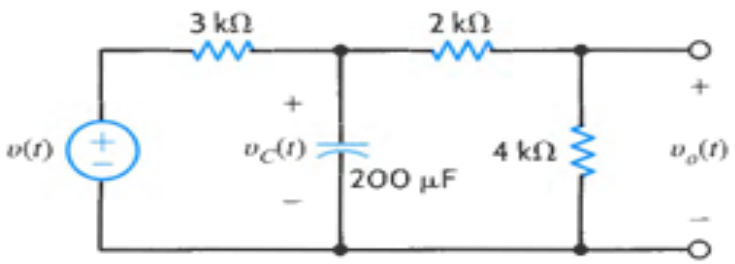
상수 전원에 대한 응답

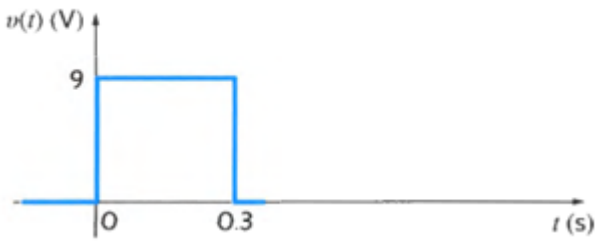
- 이 식은 상수 전원이 있는 모든 구간에서 적용된다. 상수 값들은 다를 수 있기 때문에 각각의 구간에서 따로 계산해야 한다.
- 한 구간의 끝에서의 값은 다음 구간의 초기 조건으로 사용.

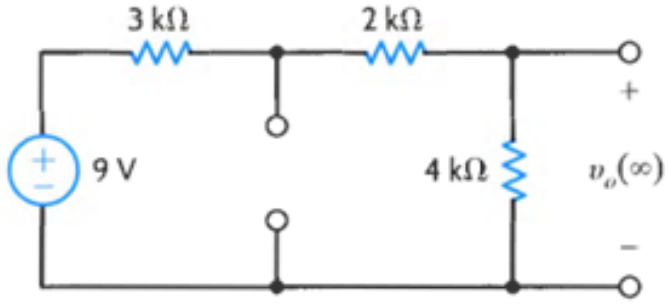
펄스 응답

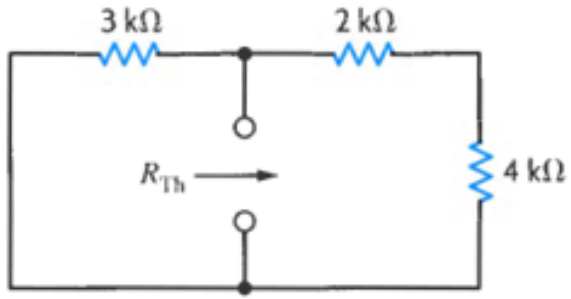
>> 학습 예제

- t>0일 때 출력전압 $v_0(t)$를 구하라.









STEP1. $v(t)$의 정상 상태 값 즉 $v_0(\infty)$ 값.

$$v_0(\infty) = \frac{9}{3k+2k+4k}4k = 4V$$

STEP2. 테브난 등가저항 구하기

$$R_{Th} = \frac{(3k)(6k)}{3k+6K} = 2k\Omega$$

시상수

$\tau = R_{Th}C$ =(2k)(200k) = 0.4s

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펄스 응답

>> 학습 예제

STEP3. 0 < t <0.3 s 일 때의 $v_0(t)$ 값

$$v_0(t) = 4\left(1 - e^{-\frac{t}{0.4}}\right)$$

STEP4. 커패시터 전압 전압분배를 사용

$$v_o(t) = \frac{2}{3} v_c(t)$$

$$v_c(t) = \frac{3}{2}\left(4 - 4e^{-t/0.4}\right) v$$

STEP5. 커패시터 전압이 연속적이므로

$$v_C(0.3-) = v_C(0.3+)$$

그러면 $v_o(0.3+) = 4\left(1 - e^{-\frac{0.3}{0.4}}\right)$

$$v_0(0.3+) = 2.11V$$

FINAL. t > 0.3 s일때 $v_0(t)$ 식

$$t > 0.3 \Rightarrow v(t) = 0$$

$$v_0(\infty) = 0 \Rightarrow K_1'' = 0$$

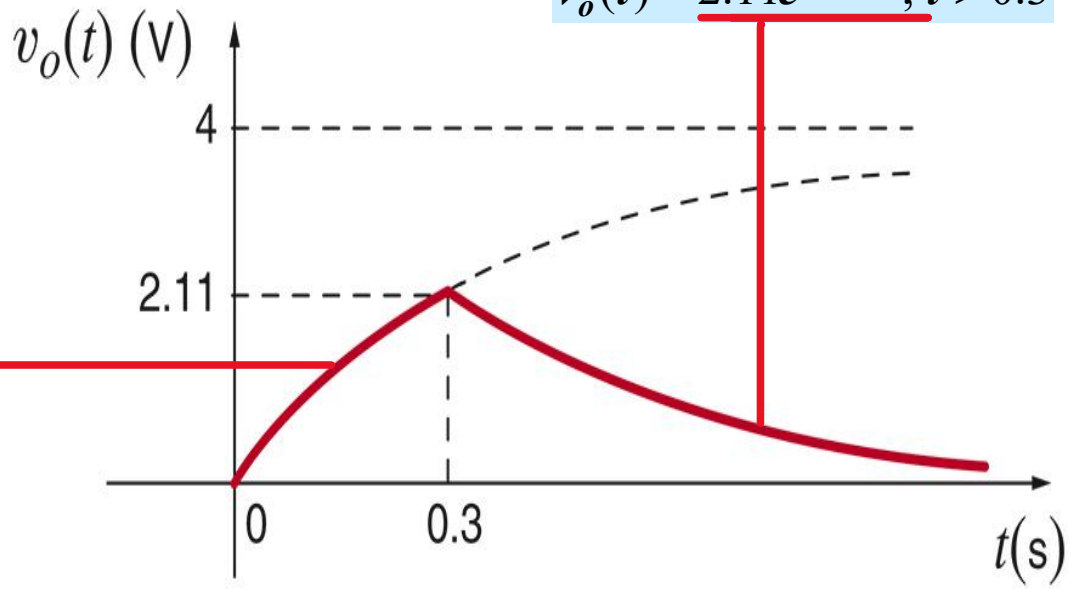
$$v_0(t) = 2.11e^{-\frac{t-0.3}{0.4}}; t > 0.3$$

펄스 응답

>> 학습 예제

$$v_o(t) = 2.11e^{-\frac{t-0.3}{0.4}}\ ;\ t > 0.3$$

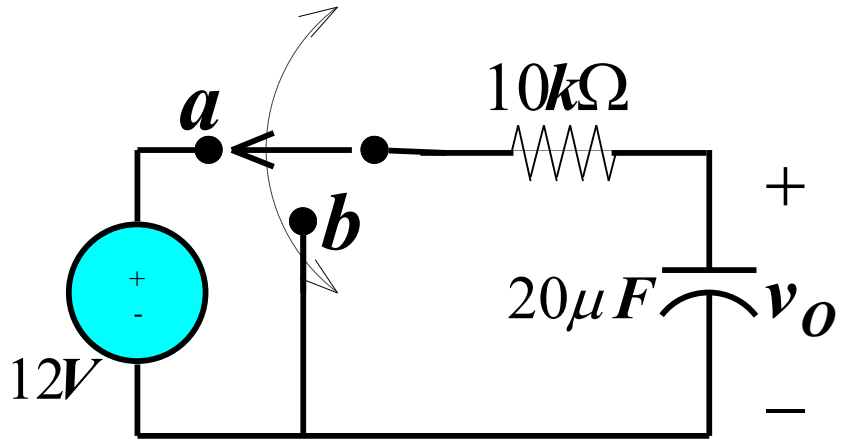
$$v_o(t) = 4\left(1 - e^{-\frac{t}{0.4}}\right)$$



펄스 응답

>> 예제

- 스위치는 초기에 a에 있다. t=0에서 b로 이동하고 t=0.5에서 다시 a로 돌아온다. t>0에 대해 $v_0(t)$를 구하라.



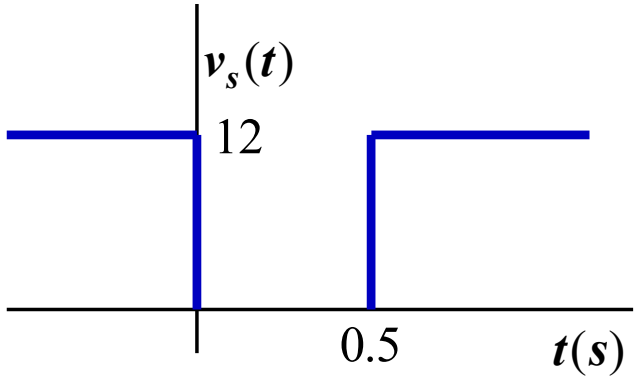
전원이 일정한 각 시간 구간마다 출력은 다음 형태를 갖는다.

$$v_0(t) = K_1 + K_2 e^{-\frac{t-t_0}{\tau}}$$

다음페이지

펄스 응답

>> 예제



시상수 구하기

$\tau = (10K\Omega)(20uF) = 0.2s$

FOR 0<t<0.5 (스위치가 b에 있을 때)

$t_0 = 0$

- 최종값

$$v_o(\infty) = 0 = K_1'$$

- 초기값

$$v(0+) = 12V = K_1' + K_2'$$

- 결론

$$v_o(t) = K_1' + K_2' e^{-\frac{t}{\tau}} \quad v_o(t) = 12e^{-\frac{t}{0.2}}, \qquad 0 < t < 0.5$$

FOR t>0.5 (스위취가 a에 있을때) $t_o = 0.5$

- t = 0.5에서의 연속조건

$$v_o(0.5+) = v_0(0.5-) = 12e^{-\frac{0.5}{0.2}} = 0.985$$

- 최종값

$$v_o(\infty) = 12 = K_1''$$

- 초기조건

$$v(0.5+) = 0.985 = K_1'' + K_2''$$

- 결론

$$K_2'' = 0.985 - 12 = -11.015$$

$$v_o(t) = K_1'' + K_2'' e^{-\frac{(t-0.5)}{\tau'}}$$

$$v_o(t) = 12 - 11.015e^{-\frac{t-0.5}{0.2}}, t > 0.5$$

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Thank you

DONGHAN KIM

HRI Lab

Human-Robot Interaction Laboratory

KYUNG HEE UNIVERSITY